

LO2 SUPPLEMENTARY GUIDE

Digital Supply Chains & Automation

Transforming operational flow through RPA, RFID, IoT sensors, and connected API ecosystems.



What is a Digital Supply Chain?

TRADITIONAL MODEL

Manual inventory counts. Phone calls and spreadsheets.
Human error and operational delay.

DIGITALIZED MODEL

Automated RFID scans. Real-time sensor telemetry.
System-to-system instant sync.



Robotic Process Automation (RPA)



Software Bots

Replicates user actions across enterprise desktop tools.



Invoice Extraction

Parses PDF documents and matches purchase orders instantly.



Rule-Based Execution

Eliminates high-volume repetitive administrative work.

RPA Exception-Based Workflow

NORMAL PATH (AUTOMATED)

Standard Orders

Valid payment detected. Automatic ledger posting and inventory adjustment.

EXCEPTION PATH (HUMAN)

Flagged Discrepancies

Mismatched invoice totals or suspicious payments routed to human staff.

ITEM IDENTIFICATION

Real-Time Inventory with RFID

10,000+

Retail items tracked instantaneously without line-of-sight scanning

Tag Detection at Gateways

Live Stock Adjustments

IoT Cold Chain Monitoring



Environmental Sensing

Continuous temp and humidity recording in pharmaceuticals.

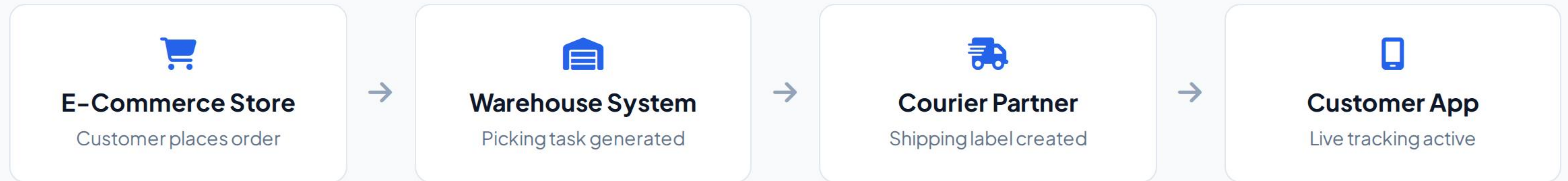
Automated Alert Thresholds

Instant alerts before product degradation occurs.

Digital Capability Matrix

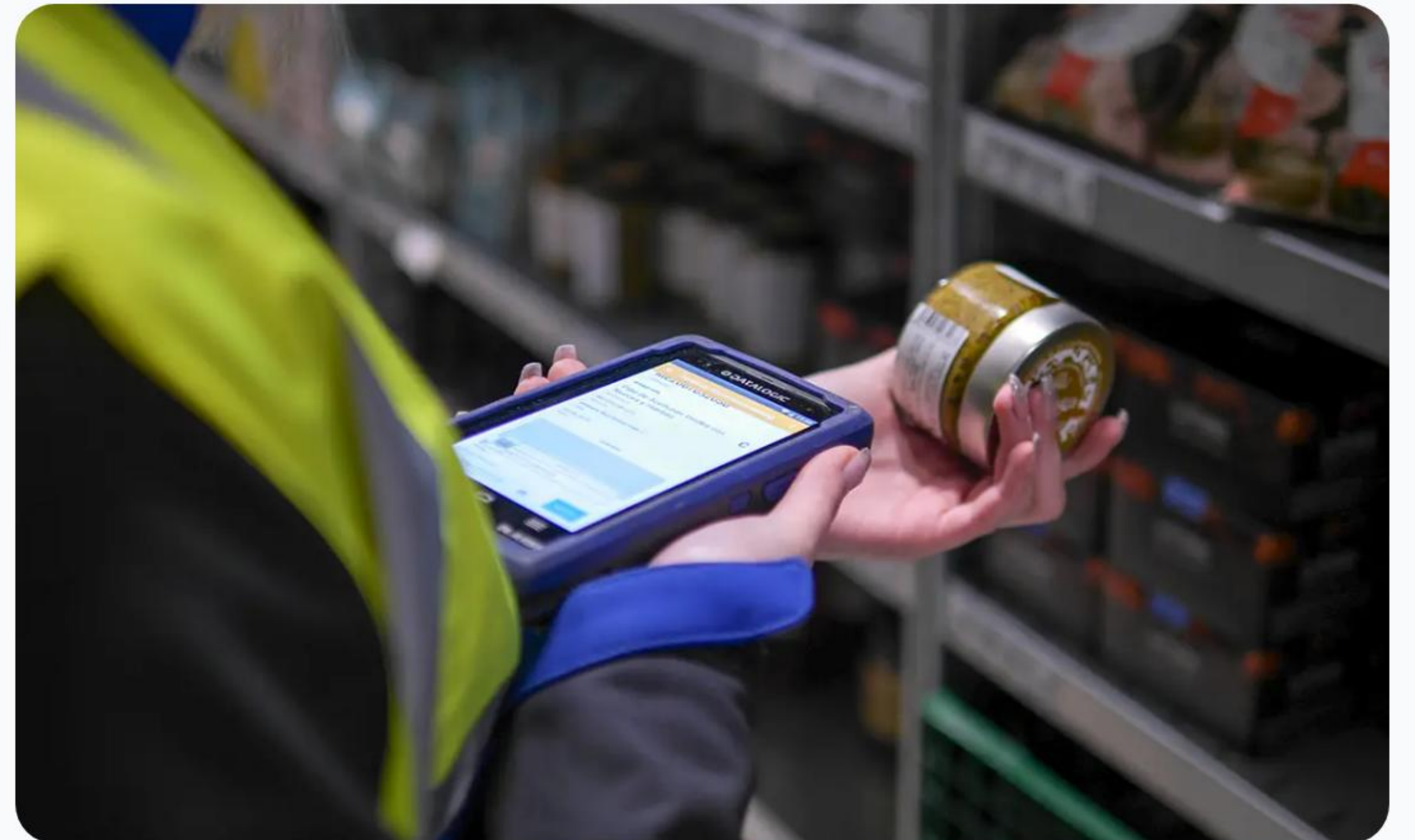
Technology	Core Function	Action Role
RPA	Automates repetitive software tasks	DO
RFID	Identifies and tracks physical goods	IDENTIFY
IoT	Captures physical world telemetry	SENSE
API	Connects disparate software platforms	CONNECT
AI	Analyzes data and predicts demand	ANALYZE

API-Connected Ecosystems



Automated Order Fulfillment Lifecycle

- 1 Payment Gateway confirms transaction via API.
- 2 RPA Bot updates stock and creates pick ticket.
- 3 RFID Scanners confirm item package contents.
- 4 Courier API generates dispatch tracking code.



End-to-End Digital Grocery Chain

API Routing

Customer order transmitted directly to fulfillment center.

RFID Bin Retrieval

Exact item slot localized in real-time.

IoT Cold Audit

Storage temperature compliance verified during pick.

Traditional vs. Digitalized Performance

MANUAL OPERATIONS

High Latency: Hours lost copying data.

Data Silos: Unsynchronized Excel files.

Blind Spots: Zero transit visibility.

DIGITAL INTEGRATION

Real-Time Sync: Sub-second updates.

Single Truth: Unified API data flow.

Total Traceability: Live sensor alerts.

SUMMARY TAKEAWAY

Connected Operations Deliver Speed and Precision

Combining RPA, RFID, IoT, and APIs eliminates manual bottlenecks
and builds resilient supply chains.



Image Sources



<https://www.supplychainbrain.com/ext/resources/2026/06/17/DIGITAL-FREIGHT-MARKETPLACE-TRUCK-iStock-Suriya-Phosri-2227640973.webp?t=1781724129>

Source: www.supplychainbrain.com



<http://www.refrigerated-trucks.com/pic/20221225125114126.jpg>

Source: www.refrigerated-trucks.com



<https://www.descartes.com/sites/default/files/media/images/2025-03/Warehouse-Worker-Scanning-Product-Barcode-1280x720px.webp>

Source: www.descartes.com